

- 5 Alice decrypts Dave's result with her private key. She subtracts the random number from Step 1 to recover the sum of everyone's salaries.
- 6 Alice divides the result by the number of people (four, in this case) and announces the result.

This way no one knew anybody else's salary and the function to calculate average salary was successfully computed. More such examples can be seen at <http://dgit.in/VO6ZU8>.

The aforementioned applications of cryptography help us understand that its use transcends almost all aspects of human dealings. Cryptography ensures security and integrity of information and prevents misuse of data by unauthorized persons. It also makes our lives convenient by providing such instruments as electronic cash and digital signatures. It was used by early man to pass on secret messages to one another, and has evolved continuously to serve our ever increasing demands. 